1 What is API security?

This chapter covers

- What is an API
- What makes an API secure or insecure
- Defining security in terms of goals
- Identifying threats and vulnerabilities
- Mechanisms for achieving security goals

Application Programming Interfaces (APIs) are everywhere. Open your smartphone or tablet and look at the apps you have installed. Almost without exception those apps are talking to one or more remote APIs to download fresh content and messages, poll for notifications, upload your new content, and perform actions on your behalf.

Load your favorite web page with the developer tools open in your browser, and you likely see dozens of API calls happening in the background to render a page that is heavily customized to you as an individual (whether you like it or not). On the server, those API calls may themselves be implemented by many *microservices*, communicating with each other via internal APIs.

Increasingly, even the everyday items in your home are talking to APIs in the cloud—from smart speakers like Amazon Echo or Google Home, to fridges, electricity meters, and lightbulbs. The *Internet of Things* (IoT) is rapidly